

● HARD

● GEOMETRY AND TRIGONOMETRY

Circles

10 questions · ~15 min

01

SAT QUESTION BANK

2026-05-29

Q1

GEOMETRY AND TRIGONOMETRY

A circle has center G , and points M and N lie on the circle. Line segments MH and NH are tangent to the circle at points M and N , respectively. If the radius of the circle is 168 millimeters and the perimeter of quadrilateral $GMHN$ is 3,856 millimeters, what is the distance, in millimeters, between points G and H ?

- A. 168
- B. 1,752
- C. 1,760
- D. 1,768

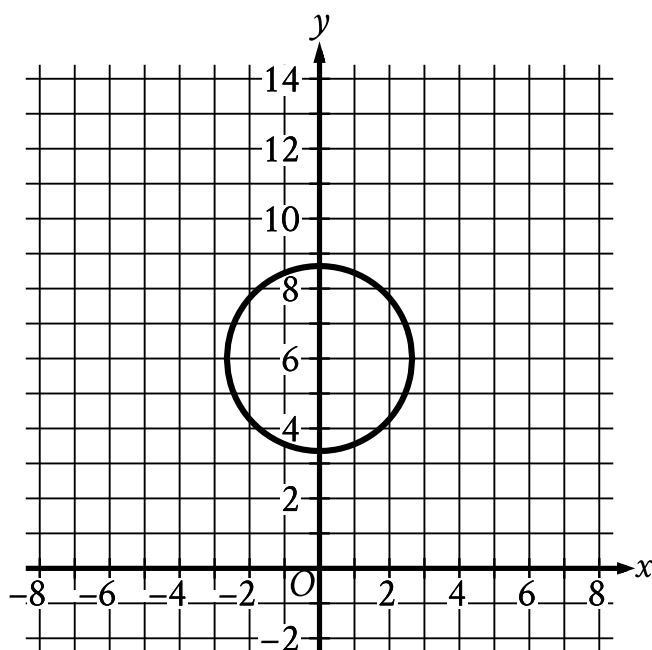
In the xy -plane, a circle has center C with coordinates h, k . Points A and B lie on the circle. Point A has coordinates $h + 1, k + \sqrt{102}$, and $\angle ACB$ is a right angle. What is the length of $\overline{AB}$?

A. $\sqrt{206}$

B. $2\sqrt{102}$

C. $103\sqrt{2}$

D. $103\sqrt{3}$



- The circle passes through the following approximate points:
 - (0 comma 8.6)
 - (2.6 comma 6)
 - (0 comma 3.4)
 - (negative 2.6 comma 6)

Circle A shown is defined by the equation $x^2 + y - 6^2 = 7$. Circle B (not shown) has the same radius but is translated 96 units to the right. If the equation of circle B is $x - h^2 + y - k^2 = a$, where h , k , and a are constants, what is the value of $4a$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

GEOMETRY AND TRIGONOMETRY

What is the value of $\sin 42\pi$?

A. 0

B. $\frac{1}{2}$

C. $\frac{\sqrt{2}}{2}$

D. 1

$$x^2 + 20x + y^2 + 16y = -20$$

The equation above defines a circle in the xy -plane. What are the coordinates of the center of the circle?

- A. $(-20, -16)$
- B. $(-10, -8)$
- C. $(10, 8)$
- D. $(20, 16)$

Q6

GEOMETRY AND TRIGONOMETRY

What is the value of $\tan \frac{92\pi}{3}$?

A. $-\sqrt{3}$

B. $-\frac{\sqrt{3}}{3}$

C. $\frac{\sqrt{3}}{3}$

D. $\sqrt{3}$

Q7

GRID-IN

GEOMETRY AND TRIGONOMETRY

Point O is the center of a circle. The measure of arc RS on this circle is 100° . What is the measure, in degrees, of its associated angle ROS ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8

GRID-IN

GEOMETRY AND TRIGONOMETRY

A circle has center O , and points R and S lie on the circle. In triangle ORS , the measure of $\angle ROS$ is 88° . What is the measure of $\angle RSO$, in degrees? (Disregard the degree symbol when entering your answer.)

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A circle in the xy -plane has its center at $(16, 17)$ and has a radius of $7k$. Which equation represents this circle?

A. $(x - 16)^2 + (y - 17)^2 = 49k$

B. $(x - 16)^2 + (y - 17)^2 = 49k^2$

C. $(x - 16)^2 + (y - 17)^2 = 7k$

D. $(x - 16)^2 + (y - 17)^2 = 7k^2$

Point F lies on a unit circle in the xy -plane and has coordinates $1, 0$. Point G is the center of the circle and has coordinates $0, 0$. Point H also lies on the circle and has coordinates $-1, y$, where y is a constant. Which of the following could be the positive measure of angle FGH , in radians?

- A. $\frac{27\pi}{2}$
- B. $\frac{29\pi}{2}$
- C. 24π
- D. 25π